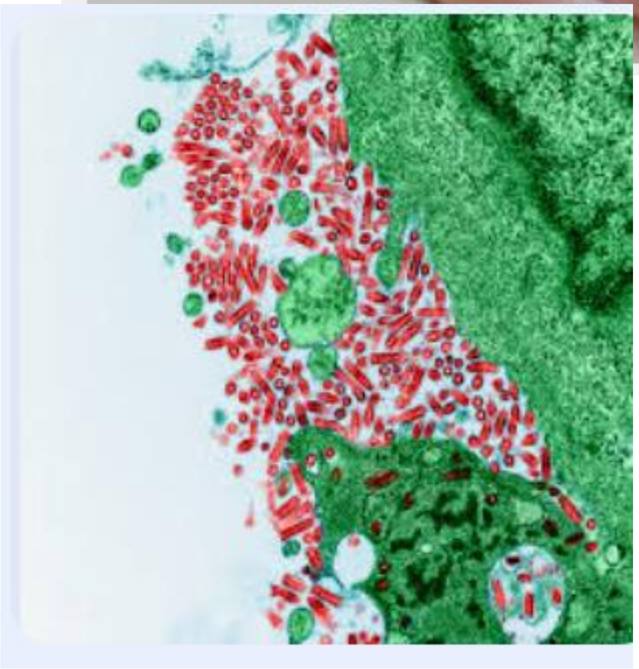
Rabies



Primary health care block

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Case presentation

- A girl aged 9 years was seen at the local hospital emergency department with a 3-day history of headache, neck pain, right arm numbness and weakness, and a temperature of 102.0°F (39.0°C), she also complained of diplopia (double vision).
- she was discharged home with a diagnosis of “muscle strain,” , after 24 hours, she came back with right arm weakness, slurred speech, diplopia, nuchal (neck) rigidity, and dysphagia (difficulty in swallowing).

The patient’s family had several pets, including cats, none of which had been ill. The parents reported that the patient had found a bat on the ground during the day at a nearby and brought it home. No other family members handled the bat, which was released the same day.

Laboratory results from the CSF revealed a WBC count of 12200 /mL (normal: 4–10) with 80% lymphocytes, a protein concentration of 96mg/dL (normal: 5–40mg/dL), and a glucose concentration of 57mg/dL (normal: 40–80mg/dL).

A CT scan of the head revealed no focal lesions.

The patient had difficulty breathing because of decreased mental status and hypersalivation. She was intubated, mechanically ventilated, and sedated because of agitation. The patient's mental status deteriorated rapidly.

Four household members and one other family member received postexposure prophylaxis (PEP) for rabies because of possible exposure to the virus through contact with the patient's saliva.

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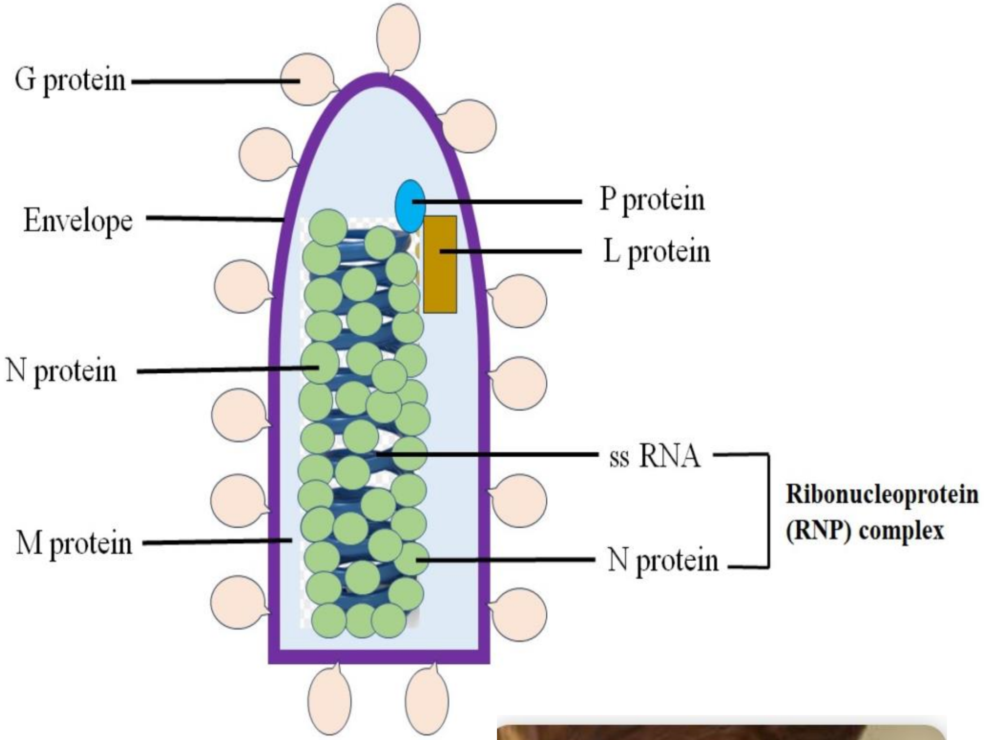


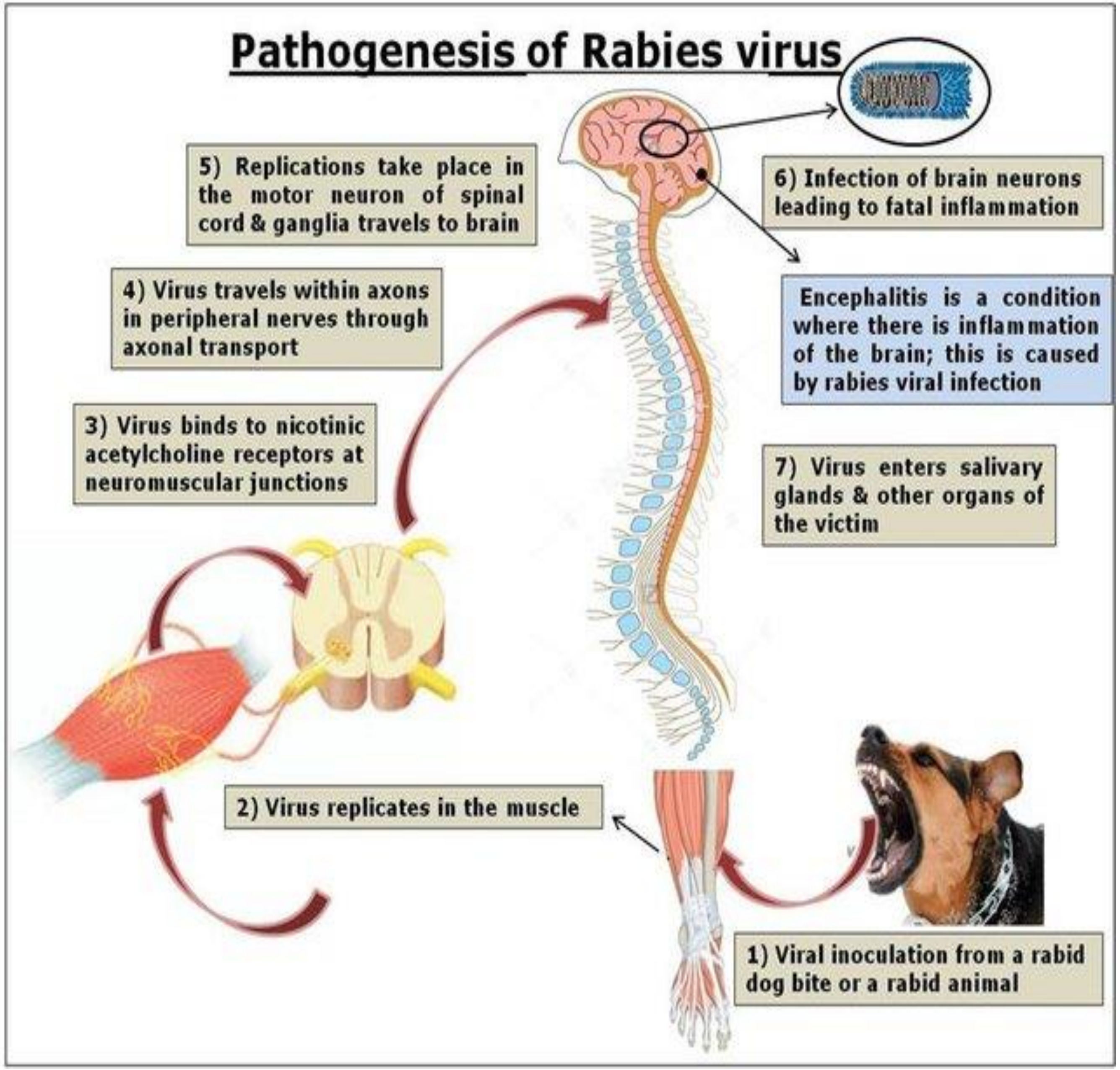
- It is a zoonotic illness of "bullet-shaped" RNA viruses with 1 to 7 serotypes belonging to the genera Lyssavirus, family Rhabdoviridae
- The term derives from Ancient Latin, referring to infection of the central nervous system (CNS), causing madness and fury.
- Animal-to-human transmission route can occur from licks, bites, or scratches.

Public health impact of rabies: Rabies virus causes an acute encephalitis in all warm-blooded hosts, including humans, and the outcome is almost always fatal. Although all species of mammals are susceptible to rabies virus infection, only a few species are important as reservoirs for the disease.

The virus multiplies in the brain, causing encephalitis & then spreads through peripheral nerves to the salivary glands & other tissues (pancreas, kidney, retina, heart).

Rabies has not been isolated from the blood of an infected person.

Rabies virus produces specific eosinophilic cytoplasmic inclusions, known as Negri bodies, in infected nerve cells. These are filled with virus nucleocapsids. The presence of such inclusions is pathognomonic for rabies (Absence in 20% of cases)



- Incubation period is prolonged and variable, average being 20-90 days (ranges from 1 week to 19 years).
- It is directly related to the distance for the virus to travel from the site of inoculation to CNS. Hence the incubation period is usually shorter in:
- 1/ Children than in adults
- 2/Bites on head, neck and upper limbs than legs
- 3/ Severe lacerations
- 4/ Presence of genetic predisposition
- 5/ Low host immunity
- 6/ High dose of inoculum
- 7/ virulence of the strain

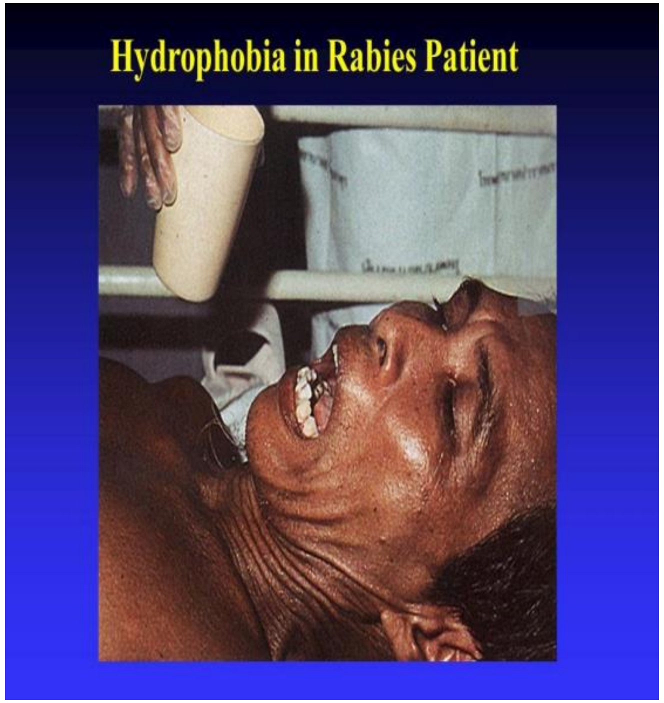
The clinical spectrum is divided into 3 phases as follows

- **Prodromal Phase** It lasts for 2-10 days, characterized by non-specific symptoms such as fever, malaise, anorexia, nausea, vomiting, photophobia, sore throat, and abnormal sensation (paresthesia, pain, or pruritus) around the wound site.

Acute Neurologic Phase: This may be either encephalitic type (80%) or paralytic type (20%).
Encephalitic or furious rabies: It lasts for 2-7 days, and is characterized by:
Hyperexcitability: Anxiety, agitation, hyperactivity, bizarre behaviour, and hallucinations may be seen
Lucid interval: The Period of hyperexcitability is typically followed by complete lucidity that becomes shorter as the disease progresses

- Autonomic (sympathetic) dysfunction features may be seen, such as lacrimation, salivation (leads to foaming at the mouth), perspiration, gooseflesh, and cardiac arrhythmia.
- Hydrophobia (fear of water) or aerophobia (fear of air)-The act of swallowing precipitates an involuntary, painful spasm of the respiratory, laryngeal, and pharyngeal muscles.
- Hydrophobia is commonly associated with furious rabies, which affects 80% of the infected people. The remaining 20% may experience a paralytic form of rabies that is marked by muscle weakness, loss of sensation, and paralysis. This form of rabies does not usually cause fear of water

These symptoms are probably due to dysfunction of infected brainstem neurons.



Paralytic or dumb rabies:

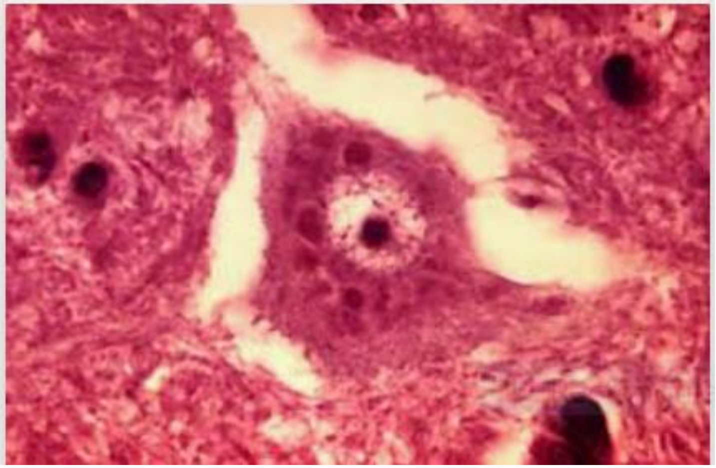
- Paralytic or dumb rabies: This occurs in 20% of cases, especially in people who are partially vaccinated or infected with the bat rabies virus. It is characterized by flaccid paralysis, which often begins in the bitten limb and progresses to quadriparesis with facial paralysis.
- However, hydrophobia and other features of encephalitic rabies are typically absent.
- Coma and Death Following the acute neurological phase, the patient develops coma that eventually leads to death within 14 days. Patients with paralytic rabies may survive longer, up to 30 days. However, death is almost certain. Recovery and survival are extremely rare.

Laboratory Diagnosis

- **Rabies Antigen Detection:** Direct immunofluorescence test (direct-IF); also called the direct fluorescent antibody (DFA) test, can be performed to detect rabies nucleoprotein antigens in specimens by using specific monoclonal antibodies tagged with fluorescent dye.
- Because of its high sensitivity and specificity, the DFA test is considered the "gold standard" method for rabies diagnosis.
- The best specimen is the hair follicle of the nape of the neck (most sensitive). A corneal impression smear can also be used. It is usually positive in the late stage, with a sensitivity of 30%
- **Antibody Detection**: Detection of CSF antibodies is more significant than serum antibodies; serum antibodies appear late and can also be present after vaccination.
- CSF antibodies appear early, and they are produced only in rabies-infected individuals, but not in response to vaccination
- Several test formats are available, which detect antibodies by employing antigens such as glycoprotein or nucleoprotein

- **Viral RNA Detection Reverse transcription-polymerase chain reaction (RT-PCR)**: can be used to amplify genes of rabies virus (e.g., genes coding for nucleoprotein or large structural protein) from brain tissue, saliva, and skin biopsy samples. It is the most sensitive and specific assay currently available for the diagnosis of rabies.

  Negri Body Detection. It is useful for postmortem diagnosis of rabies. It is an intracytoplasmic eosinophilic inclusion with characteristic basophilic inner granules. It is sharply demarcated, spherical to oval, and about 2-10 μm in size; surrounded by a halo. Most common sites of Negri bodies are neurons of the cerebellum and hippocampus; however, they can also be less frequently seen in cortical and brainstem neurons Negri body detection is pathognomonic of rabies. However, it may not be detected in 20% of cases. Therefore, the absence of Negri bodies does not rule out the diagnosis of rabies.



Treatment

1/ There is no specific treatment for rabies.

2/ Symptomatic treatment may prolong life, but the outcome is almost always fatal.

3/ Local Wound Care:

It can greatly decrease the risk of rabies if initiated immediately. Even if the patient reports late, care must still be performed, as the rabies virus is known to persist in the skin for an extended period.

A/ Physical cleansing: All bite wounds and scratches should be washed thoroughly with soap and water for 15 minutes

B/ Chemical inactivation: Antiseptics such as povidone iodine or alcohol can be used to inactivate the residual viruses

4/ Isolation: Patient should be isolated in a quiet room, protected as far as possible from external stimuli such as bright light, noise, water, or cold air, which can precipitate spasms.

5/ Sedatives and anti-anxiety drugs, such as morphine, can be used.

6/ Hydration and urination should be properly maintained.

7/ Biological neutralization of the virus by giving anti-rabies immunoglobulin:

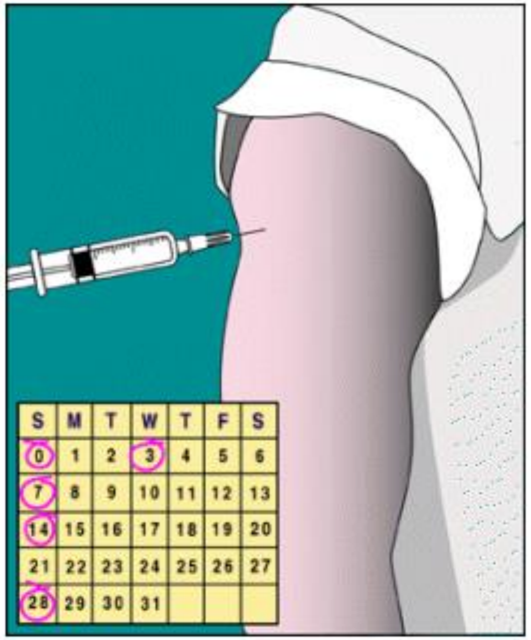
The action of passively administered anti-rabies antibodies is to neutralize some of the inoculated virus & to lower its concentration in the body, providing additional time for a vaccine to stimulate active antibody production to prevent entry to the CNS.

Passive Immunization (Rabies immunoglobulin) Rabies immunoglobulins (RIG) have the property of binding with the rabies virus, thereby resulting in neutralization and thus loss of infectivity of the virus. Hence, RIGs are usually administered locally at the site of exposure.

Two types of RI Gs are available:
1/ Equine rabies immunoglobulin (ERIG):
2/ Human rabies immunoglobulin (HRIG): It is given in a dose of 20 IU /kg.
It is devoid of side effects. Maximum volume of RIGs should be infiltrated into and around the bite wound(s); remaining volume, if any, should be administered by deep intramuscular injection at a site distant from the vaccine injection site , RIG should be administered within 24 hours; maximum up to the 7th day.

8/Debridement of devitalized tissues, Tetanus prophylaxis should be given Antibiotic treatment is initiated to prevent secondary bacterial infection Suturing is contraindicated: Bite wounds should not be immediately sutured, as it may help in spreading of the virus into deeper tissues Do not touch the wound(s) with bare hands Do not apply any materials

- **Prognosis**: Mortality in rabies is almost 100%; however, it is preventable by administration of post-exposure therapy during the early incubation period.
- Prevention:
- There is no antiviral treatment for rabies after symptoms of the disease appear. However, there is a safe and extremely effective new rabies vaccine regimen that provides immunity to rabies when administered after an exposure. Preexposure prophylaxis: For persons at special risk of exposure to rabies. It consists of immunization with either HDCV or RAV vaccine[1.0 ml, IM (deltoid) at 0,7,21, or 28 days with a booster dose at day 0 (depending on the exposure risk category].



Epidemiology